

## COMPLETE LEARNING SESSION

# Landslides, Avalanches and GLOF Risk - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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# Landslides, Avalanches and GLOF Risk - Learner-v2

## Complete Learning Session

Authoring-only generation: 2026-09-04. No PDF was rendered and no tracker or index was mutated.

### SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT

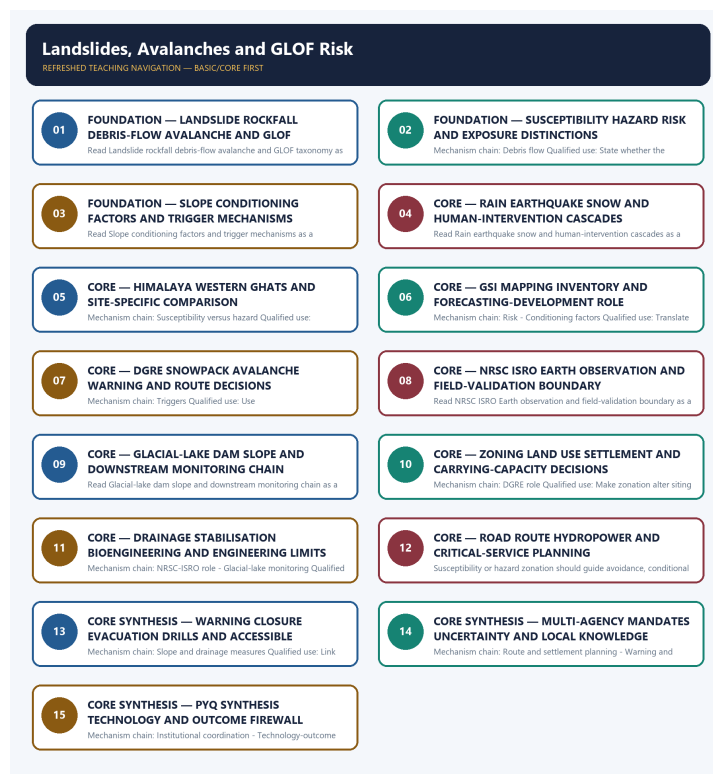
- **Generation date:** 2026-09-04.
- **Repository-first evidence:** the Basic owner is taught first and preserved in full; the Advanced owner is retained only in the optional final teaching block.
- **OCR evidence:** Repository Markdown was primary. OCR-searchable official UPSC papers were supplementary only for printed routed demands. No answer key, warning performance, notification effect, implementation outcome, casualty reduction or disaster metric was inferred.
- **Qdrant:** not used; repository Markdown and available OCR context were sufficient.
- **PYQ integrity:** All three cards are verified direct routes: 2019 landslide zonation, 2021 Himalayan-versus-Western-Ghats causes, and 2021 landslide causes/effects/National Landslide Risk Management Strategy. Avalanche and GLOF content is not misrepresented as printed in those stems.
- **Live-link boundary:** Official attempts covered GSI Bhusanket roles, a dated DGRE avalanche bulletin, NRSC's Landslide Atlas, ISRO's atlas route and NDMA pages. Thin, raw-PDF and transport-failed pages are logged; no current hazard level, forecast accuracy, lake count, route status, casualty, event attribution or operational-readiness claim was invented.
- **Fact/inference discipline:** no current-affairs item, PYQ wording, figure or quotation is invented.

### LIVE OFFICIAL-SOURCE ATTEMPT LOG

The checks below were made on 2026-09-04. Substantive official text is used only for the proposition it supports; title-only, blocked or thin pages are recorded and supply no factual claim.

- <https://bhusanket.gsi.gov.in/rolesAndResponsibility.html> - fetched 2026-09-04; GSI described landslide susceptibility work, hazard/risk research, early-warning development and an experimental regional rainfall-threshold programme. <https://ndma.gov.in/Natural-Hazards/Landslide> - attempted 2026-09-04; the NDMA page failed at the transport layer. No national service-coverage claim was inferred.
- [https://www.drdo.gov.in/drdo/sites/default/files/avalanche\\_warning\\_bulletin/DGRE\\_AWB\\_30-Apr-2026.pdf](https://www.drdo.gov.in/drdo/sites/default/files/avalanche_warning_bulletin/DGRE_AWB_30-Apr-2026.pdf) - fetched 2026-09-04 as raw official PDF content, confirming a dated DGRE avalanche-warning bulletin exists. <https://ndrf.gov.in/en/about-us> - fetched 2026-09-04 only for the general specialised-response role. No bulletin level, route safety or current readiness was imported.
- [https://www.nrsc.gov.in/nrscnew/resources\\_atlas\\_landslide.php](https://www.nrsc.gov.in/nrscnew/resources_atlas_landslide.php) - fetched 2026-09-04; NRSC described its 1998-2022 inventory, seasonal, event-based and route-wise classes, and satellite/field-validation inputs. The figures remain dated atlas metadata, not a forecast or current loss estimate.
- [https://www.isro.gov.in/Landslide\\_Atlas\\_India.html](https://www.isro.gov.in/Landslide_Atlas_India.html) - fetched 2026-09-04; the official page returned only a thin atlas title. <https://cwc.gov.in/> - fetched 2026-09-04 for CWC's general water-resources coordination remit, not a GLOF-warning-performance claim. <https://ndma.gov.in/sites/default/files/PDF/Guidelines/Guidelines-on-Management-of-Glacial-Lake-Outburst-Floods.pdf> - searched 2026-09-04; no retrievable text was obtained. GLOF measures therefore remain bounded to the canonical owner.
- <https://nidm.gov.in/modules.asp> - searched 2026-09-04; the official NIDM modules route was identified but supplied no hazard-specific operational status. It was retained only as a capacity-building source.

## BASIC LEARNING SESSION



Refreshed teaching navigation

*Distinct embedded teaching-navigation image. The separate continuous Cārvāka-style flowchart package remains an independent at-a-glance artifact.*

## SESSION 1 - FOUNDATION - LANDSLIDE ROCKFALL DEBRIS-FLOW AVALANCHE AND GLOF TAXONOMY

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** The decisive boundary is Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms.

**Technical definition:** Read Landslide rockfall debris-flow avalanche and GLOF taxonomy as a sequence: identify Landslide, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Landslide rockfall debris-flow avalanche and GLOF taxonomy as a sequence: identify Landslide, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- FOUNDATION
- Landslide rockfall debris-flow avalanche
- GLOF taxonomy
- Mechanism chain
- Qualified use
- A landslide

**How to use them:** Frame the answer through FOUNDATION; define Landslide rockfall debris-flow avalanche, connect GLOF taxonomy with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

## VISUAL FIRST

LANDSLIDE ROCKFALL DEBRIS-FLOW AVALANCHE AND GLOF TAXONOMY

01. Landslide

|  
v

02. Rockfall

STATUS / CATEGORY FIREWALL -> Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow.

## SIMPLE SYSTEM EXAMPLE

Read Landslide rockfall debris-flow avalanche and GLOF taxonomy as a sequence: identify Landslide, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms. This keeps Landslide and Rockfall in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.
- Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow.

## EXAMINER CAUTION

- Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Define the moving material and process before discussing management.

## MINI RECAP

- **Mechanism chain:** Landslide -> Rockfall
- **Qualified use:** Define the moving material and process before discussing management.

## CLOSING RECALL FLOW - FOUNDATION - LANDSLIDE ROCKFALL DEBRIS-FLOW AVALANCHE AND GLOF TAXONOMY

START / CONCEPT: FOUNDATION - Landslide rockfall debris-flow avalanche and GLOF taxonomy  
|  
v  
EXACT TERMS: FOUNDATION · Landslide rockfall debris-flow avalanche · GLOF taxonomy · Mechanism chain · Qualified use · A landslide  
|  
v  
MECHANISM / ARGUMENT: Mechanism chain: Landslide - Rockfall Qualified use: Define the moving material and process before discussing management.  
|  
v  
CONSEQUENCE / CONTRAST: The decisive boundary is Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms.  
|  
v  
UPSC TRAP / ANSWER-USE: Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms.  
|  
v  
ANSWER-GRABBING FORMULATION: Read Landslide rockfall debris-flow avalanche and GLOF taxonomy as a sequence: identify Landslide, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## SESSION 2 - FOUNDATION - SUSCEPTIBILITY HAZARD RISK AND EXPOSURE DISTINCTIONS

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Susceptibility hazard risk and exposure distinctions as a sequence: identify Debris flow, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Debris flow Qualified use: State whether the evidence answers where, when or who may be harmed.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Susceptibility hazard risk and exposure distinctions as a sequence: identify Debris flow, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- FOUNDATION
- Susceptibility hazard risk
- exposure distinctions
- Mechanism chain
- Qualified use
- A debris flow

**How to use them:** Frame the answer through FOUNDATION; define Susceptibility hazard risk, connect exposure distinctions with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

## VISUAL FIRST

### SUSCEPTIBILITY HAZARD RISK AND EXPOSURE DISTINCTIONS

01. Debris flow

STATUS / CATEGORY FIREWALL -> Do not confuse intense rainfall or a cloudburst with the mass movement it may trigger.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

## SIMPLE SYSTEM EXAMPLE

Read Susceptibility hazard risk and exposure distinctions as a sequence: identify Debris flow, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not confuse intense rainfall or a cloudburst with the mass movement it may trigger. This keeps Debris flow in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

## EXAMINER CAUTION

- Do not confuse intense rainfall or a cloudburst with the mass movement it may trigger.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** State whether the evidence answers where, when or who may be harmed.

## MINI RECAP

- **Mechanism chain:** Debris flow
- **Qualified use:** State whether the evidence answers where, when or who may be harmed.

## CLOSING RECALL FLOW - FOUNDATION - SUSCEPTIBILITY HAZARD RISK AND EXPOSURE DISTINCTIONS

START / CONCEPT: FOUNDATION - Susceptibility hazard risk and exposure distinctions

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EXACT TERMS: FOUNDATION · Susceptibility hazard risk · exposure distinctions · Mechanism chain · Qualified use · A debris flow

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Debris flow Qualified use: State whether the evidence answers where, when or who may be harmed.

|  
v

CONSEQUENCE / CONTRAST: A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

|  
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not confuse intense rainfall or a cloudburst with the mass movement it may trigger.

|  
v

ANSWER-GRABBING FORMULATION: Read Susceptibility hazard risk and exposure distinctions as a sequence: identify Debris flow, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

# SESSION 3 - FOUNDATION - SLOPE CONDITIONING FACTORS AND TRIGGER MECHANISMS

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.

**Technical definition:** Read Slope conditioning factors and trigger mechanisms as a sequence: identify Avalanche, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Slope conditioning factors and trigger mechanisms as a sequence: identify Avalanche, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Slope conditioning factors**
- **trigger mechanisms**
- **Mechanism chain**
- **Qualified use**
- **An avalanche**

**How to use them:** Frame the answer through FOUNDATION; define Slope conditioning factors, connect trigger mechanisms with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

## VISUAL FIRST

SLOPE CONDITIONING FACTORS AND TRIGGER MECHANISMS

01. Avalanche

STATUS / CATEGORY FIREWALL -> Do not equate susceptibility, hazard and risk maps.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.

## SIMPLE SYSTEM EXAMPLE

Read Slope conditioning factors and trigger mechanisms as a sequence: identify Avalanche, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not equate susceptibility, hazard and risk maps. This keeps Avalanche in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.

## EXAMINER CAUTION

- Do not equate susceptibility, hazard and risk maps.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Separate persistent conditions from the event trigger.

## MINI RECAP

- **Mechanism chain:** Avalanche
- **Qualified use:** Separate persistent conditions from the event trigger.

## CLOSING RECALL FLOW - FOUNDATION - SLOPE CONDITIONING FACTORS AND TRIGGER MECHANISMS

START / CONCEPT: FOUNDATION - Slope conditioning factors and trigger mechanisms

|  
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EXACT TERMS: FOUNDATION · Slope conditioning factors · trigger mechanisms · Mechanism chain ·  
Qualified use · An avalanche

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Avalanche Qualified use: Separate persistent conditions from  
the event trigger.

|  
v

CONSEQUENCE / CONTRAST: An avalanche is rapid downslope movement of snow, ice and entrained  
material produced by snowpack instability, slope and weather conditions; it is not a landslide  
label for every mountain failure.

|  
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not equate susceptibility, hazard and risk maps.

|  
v

ANSWER-GRABBING FORMULATION: Read Slope conditioning factors and trigger mechanisms as a sequence:  
identify Avalanche, follow the named mechanism to its institutional response, and stop at the last  
verified status rather than assuming the next stage.

## SESSION 4 - CORE - RAIN EARTHQUAKE SNOW AND HUMAN-INTERVENTION CASCADES

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

**Technical definition:** Read Rain earthquake snow and human-intervention cascades as a sequence: identify GLOF, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Rain earthquake snow and human-intervention cascades as a sequence: identify GLOF, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- Rain earthquake snow
- human-intervention cascades
- Mechanism chain
- Qualified use
- A Glacial Lake Outburst Flood
- Glacial Lake Outburst Flood

**How to use them:** Frame the answer through Rain earthquake snow; define human-intervention cascades, connect Mechanism chain with Qualified use to explain the mechanism, and use A Glacial Lake Outburst Flood for the decisive comparison or qualification.

## VISUAL FIRST

RAIN EARTHQUAKE SNOW AND HUMAN-INTERVENTION CASCADES

01. GLOF

STATUS / CATEGORY FIREWALL -> Do not assign GSI's landslide role to DGRE or DGRE's avalanche role to ISRO.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

## SIMPLE SYSTEM EXAMPLE

Read Rain earthquake snow and human-intervention cascades as a sequence: identify GLOF, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not assign GSI's landslide role to DGRE or DGRE's avalanche role to ISRO. This keeps GLOF in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

## EXAMINER CAUTION

- Do not assign GSI's landslide role to DGRE or DGRE's avalanche role to ISRO.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Trace the cascade without claiming every trigger causes every outcome.

## MINI RECAP

- **Mechanism chain:** GLOF
- **Qualified use:** Trace the cascade without claiming every trigger causes every outcome.

## CLOSING RECALL FLOW - CORE - RAIN EARTHQUAKE SNOW AND HUMAN-INTERVENTION CASCADES

START / CONCEPT: CORE - Rain earthquake snow and human-intervention cascades

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EXACT TERMS: Rain earthquake snow • human-intervention cascades • Mechanism chain • Qualified use •

A Glacial Lake Outburst Flood • Glacial Lake Outburst Flood

|

v

MECHANISM / ARGUMENT: Mechanism chain: GLOF Qualified use: Trace the cascade without claiming every trigger causes every outcome.

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CONSEQUENCE / CONTRAST: The decisive boundary is Do not assign GSI's landslide role to DGRE or DGRE's avalanche role to ISRO.

|

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UPSC TRAP / ANSWER-USE: Do not assign GSI's landslide role to DGRE or DGRE's avalanche role to ISRO.

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ANSWER-GRABBING FORMULATION: Read Rain earthquake snow and human-intervention cascades as a sequence: identify GLOF, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## SESSION 5 - CORE - HIMALAYA WESTERN GHATS AND SITE-SPECIFIC COMPARISON

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Himalaya Western Ghats and site-specific comparison as a sequence: identify Susceptibility versus hazard, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Susceptibility versus hazard Qualified use: Compare geology rainfall relief and intervention rather than repeating one list.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Himalaya Western Ghats and site-specific comparison as a sequence: identify Susceptibility versus hazard, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- Himalaya Western Ghats
- site-specific comparison
- Mechanism chain
- Qualified use
- Susceptibility indicates where terrain
- Susceptibility

**How to use them:** Frame the answer through Himalaya Western Ghats; define site-specific comparison, connect Mechanism chain with Qualified use to explain the mechanism, and use Susceptibility indicates where terrain for the decisive comparison or qualification.

### VISUAL FIRST

HIMALAYA WESTERN GHATS AND SITE-SPECIFIC COMPARISON

01. Susceptibility versus hazard

STATUS / CATEGORY FIREWALL -> Do not convert experimental forecasting into nationwide operational warning.

*The rail fixes the institutional sequence and claim boundary before explanation.*

### CORE EXPLANATION

Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

### SIMPLE SYSTEM EXAMPLE

Read Himalaya Western Ghats and site-specific comparison as a sequence: identify Susceptibility versus hazard, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### TECHNICAL DISTINCTION

The decisive boundary is Do not convert experimental forecasting into nationwide operational warning. This keeps Susceptibility versus hazard in the correct risk category, institutional mandate and evidence status.

### NAMED EVIDENCE AND MECHANISM

- Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

### EXAMINER CAUTION

- Do not convert experimental forecasting into nationwide operational warning.

### EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Compare geology rainfall relief and intervention rather than repeating one list.

### MINI RECAP

- **Mechanism chain:** Susceptibility versus hazard
- **Qualified use:** Compare geology rainfall relief and intervention rather than repeating one list.

### CLOSING RECALL FLOW - CORE - HIMALAYA WESTERN GHATS AND SITE-SPECIFIC COMPARISON

START / CONCEPT: CORE - Himalaya Western Ghats and site-specific comparison

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EXACT TERMS: Himalaya Western Ghats · site-specific comparison · Mechanism chain · Qualified use · Susceptibility indicates where terrain · Susceptibility

|

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MECHANISM / ARGUMENT: Mechanism chain: Susceptibility versus hazard Qualified use: Compare geology rainfall relief and intervention rather than repeating one list.

|

v

CONSEQUENCE / CONTRAST: Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

|

v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not convert experimental forecasting into nationwide operational warning.

|

v

ANSWER-GRABBING FORMULATION: Read Himalaya Western Ghats and site-specific comparison as a sequence: identify Susceptibility versus hazard, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## SESSION 6 - CORE - GSI MAPPING INVENTORY AND FORECASTING-DEVELOPMENT ROLE

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read GSI mapping inventory and forecasting-development role as a sequence: identify Risk, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Risk - Conditioning factors Qualified use: Translate a map or inventory into a planning decision and preserve service status.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read GSI mapping inventory and forecasting-development role as a sequence: identify Risk, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- **GSI mapping inventory**
- **forecasting-development role**
- **Mechanism chain**
- **Qualified use**
- **Risk**
- **Read GSI**

**How to use them:** Frame the answer through GSI mapping inventory; define forecasting-development role, connect Mechanism chain with Qualified use to explain the mechanism, and use Risk for the decisive comparison or qualification.

## VISUAL FIRST

GSI MAPPING INVENTORY AND FORECASTING-DEVELOPMENT ROLE

01. Risk

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v

02. Conditioning factors

STATUS / CATEGORY FIREWALL -> Do not describe remote sensing as direct prediction of every failure.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

## SIMPLE SYSTEM EXAMPLE

Read GSI mapping inventory and forecasting-development role as a sequence: identify Risk, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not describe remote sensing as direct prediction of every failure. This keeps Risk and Conditioning factors in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.
- Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

## EXAMINER CAUTION

- Do not describe remote sensing as direct prediction of every failure.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Translate a map or inventory into a planning decision and preserve service status.

## MINI RECAP

- **Mechanism chain:** Risk -> Conditioning factors
- **Qualified use:** Translate a map or inventory into a planning decision and preserve service status.

## CLOSING RECALL FLOW - CORE - GSI MAPPING INVENTORY AND FORECASTING-DEVELOPMENT ROLE

START / CONCEPT: CORE - GSI mapping inventory and forecasting-development role

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EXACT TERMS: GSI mapping inventory · forecasting-development role · Mechanism chain · Qualified use  
· Risk · Read GSI

|  
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MECHANISM / ARGUMENT: Mechanism chain: Risk - Conditioning factors Qualified use: Translate a map or inventory into a planning decision and preserve service status.

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CONSEQUENCE / CONTRAST: The decisive boundary is Do not describe remote sensing as direct prediction of every failure.

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UPSC TRAP / ANSWER-USE: Do not describe remote sensing as direct prediction of every failure.

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ANSWER-GRABBING FORMULATION: Read GSI mapping inventory and forecasting-development role as a sequence: identify Risk, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## SESSION 7 - CORE - DGRE SNOWPACK AVALANCHE WARNING AND ROUTE DECISIONS

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read DGRE snowpack avalanche warning and route decisions as a sequence: identify Triggers, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Triggers Qualified use: Use snowpack-slope-weather data for avalanche decisions only.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read DGRE snowpack avalanche warning and route decisions as a sequence: identify Triggers, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- **DGRE snowpack avalanche warning**
- **route decisions**
- **Mechanism chain**
- **Qualified use**
- **Intense**
- **GLOF**

**How to use them:** Frame the answer through DGRE snowpack avalanche warning; define route decisions, connect Mechanism chain with Qualified use to explain the mechanism, and use Intense for the decisive comparison or qualification.

## VISUAL FIRST

DGRE SNOWPACK AVALANCHE WARNING AND ROUTE DECISIONS

01. Triggers

STATUS / CATEGORY FIREWALL -> Do not assume retaining walls or drainage make an unsuitable site safe.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.

## SIMPLE SYSTEM EXAMPLE

Read DGRE snowpack avalanche warning and route decisions as a sequence: identify Triggers, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not assume retaining walls or drainage make an unsuitable site safe. This keeps Triggers in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.

## EXAMINER CAUTION

- Do not assume retaining walls or drainage make an unsuitable site safe.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Use snowpack-slope-weather data for avalanche decisions only.

## MINI RECAP

- **Mechanism chain:** Triggers
- **Qualified use:** Use snowpack-slope-weather data for avalanche decisions only.

## CLOSING RECALL FLOW - CORE - DGRE SNOWPACK AVALANCHE WARNING AND ROUTE DECISIONS

START / CONCEPT: CORE - DGRE snowpack avalanche warning and route decisions

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EXACT TERMS: DGRE snowpack avalanche warning · route decisions · Mechanism chain · Qualified use · Intense · GLOF

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MECHANISM / ARGUMENT: Mechanism chain: Triggers Qualified use: Use snowpack-slope-weather data for avalanche decisions only.

|

v

CONSEQUENCE / CONTRAST: Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.

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UPSC TRAP / ANSWER-USE: The decisive boundary is Do not assume retaining walls or drainage make an unsuitable site safe.

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ANSWER-GRABBING FORMULATION: Read DGRE snowpack avalanche warning and route decisions as a sequence: identify Triggers, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## SESSION 8 - CORE - NRSC ISRO EARTH OBSERVATION AND FIELD-VALIDATION BOUNDARY

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** The decisive boundary is Do not recommend road closure or evacuation without alternative access and safe destinations.

**Technical definition:** Read NRSC ISRO Earth observation and field-validation boundary as a sequence: identify Cascading risk, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read NRSC ISRO Earth observation and field-validation boundary as a sequence: identify Cascading risk, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- NRSC ISRO Earth observation
- field-validation boundary
- Mechanism chain
- Qualified use
- Earthquake
- Read NRSC ISRO Earth

**How to use them:** Frame the answer through NRSC ISRO Earth observation; define field-validation boundary, connect Mechanism chain with Qualified use to explain the mechanism, and use Earthquake for the decisive comparison or qualification.

### VISUAL FIRST

NRSC ISRO EARTH OBSERVATION AND FIELD-VALIDATION BOUNDARY

01. Cascading risk

STATUS / CATEGORY FIREWALL -> Do not recommend road closure or evacuation without alternative access and safe destinations.

*The rail fixes the institutional sequence and claim boundary before explanation.*

### CORE EXPLANATION

Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.

### SIMPLE SYSTEM EXAMPLE

Read NRSC ISRO Earth observation and field-validation boundary as a sequence: identify Cascading risk, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### TECHNICAL DISTINCTION

The decisive boundary is Do not recommend road closure or evacuation without alternative access and safe destinations. This keeps Cascading risk in the correct risk category, institutional mandate and evidence status.

### NAMED EVIDENCE AND MECHANISM

- Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.

### EXAMINER CAUTION

- Do not recommend road closure or evacuation without alternative access and safe destinations.

### EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Treat satellite products as screening and decision support requiring validation.

### MINI RECAP

- **Mechanism chain:** Cascading risk
- **Qualified use:** Treat satellite products as screening and decision support requiring validation.

### CLOSING RECALL FLOW - CORE - NRSC ISRO EARTH OBSERVATION AND FIELD-VALIDATION BOUNDARY

START / CONCEPT: CORE - NRSC ISRO Earth observation and field-validation boundary

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EXACT TERMS: NRSC ISRO Earth observation · field-validation boundary · Mechanism chain · Qualified use · Earthquake · Read NRSC ISRO Earth

|  
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MECHANISM / ARGUMENT: Mechanism chain: Cascading risk Qualified use: Treat satellite products as screening and decision support requiring validation.

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CONSEQUENCE / CONTRAST: The decisive boundary is Do not recommend road closure or evacuation without alternative access and safe destinations.

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UPSC TRAP / ANSWER-USE: Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.

|  
v

ANSWER-GRABBING FORMULATION: Read NRSC ISRO Earth observation and field-validation boundary as a sequence: identify Cascading risk, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## SESSION 9 - CORE - GLACIAL-LAKE DAM SLOPE AND DOWNSTREAM MONITORING CHAIN

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** The decisive boundary is Do not use a dynamic glacial-lake list or programme figure without its date and owner.

**Technical definition:** Read Glacial-lake dam slope and downstream monitoring chain as a sequence: identify GSI role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Glacial-lake dam slope and downstream monitoring chain as a sequence: identify GSI role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- **Glacial-lake dam slope**
- **downstream monitoring chain**
- **Mechanism chain**
- **Qualified use**
- **GSI**
- **Bhusanket**

**How to use them:** Frame the answer through Glacial-lake dam slope; define downstream monitoring chain, connect Mechanism chain with Qualified use to explain the mechanism, and use GSI for the decisive comparison or

qualification.

## VISUAL FIRST

GLACIAL-LAKE DAM SLOPE AND DOWNSTREAM MONITORING CHAIN

01. GSI role

STATUS / CATEGORY FIREWALL -> Do not use a dynamic glacial-lake list or programme figure without its date and owner.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

## SIMPLE SYSTEM EXAMPLE

Read Glacial-lake dam slope and downstream monitoring chain as a sequence: identify GSI role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not use a dynamic glacial-lake list or programme figure without its date and owner. This keeps GSI role in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

## EXAMINER CAUTION

- Do not use a dynamic glacial-lake list or programme figure without its date and owner.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Connect lake and dam change to downstream exposure and evacuation.

## MINI RECAP

- **Mechanism chain:** GSI role
- **Qualified use:** Connect lake and dam change to downstream exposure and evacuation.

## CLOSING RECALL FLOW - CORE - GLACIAL-LAKE DAM SLOPE AND DOWNSTREAM MONITORING CHAIN

START / CONCEPT: CORE - Glacial-lake dam slope and downstream monitoring chain

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EXACT TERMS: Glacial-lake dam slope · downstream monitoring chain · Mechanism chain · Qualified use  
· GSI · Bhusanket

|

v

MECHANISM / ARGUMENT: Mechanism chain: GSI role Qualified use: Connect lake and dam change to downstream exposure and evacuation.

|

v

CONSEQUENCE / CONTRAST: The decisive boundary is Do not use a dynamic glacial-lake list or programme figure without its date and owner.

|

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UPSC TRAP / ANSWER-USE: Do not use a dynamic glacial-lake list or programme figure without its date and owner.

|

v

ANSWER-GRABBING FORMULATION: Read Glacial-lake dam slope and downstream monitoring chain as a sequence: identify GSI role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

# SESSION 10 - CORE - ZONING LAND USE SETTLEMENT AND CARRYING-CAPACITY DECISIONS

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Zoning land use settlement and carrying-capacity decisions as a sequence: identify DGRE role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: DGRE role Qualified use: Make zonation alter siting construction and route choices.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Zoning land use settlement and carrying-capacity decisions as a sequence: identify DGRE role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## MUST-WRITE KEYWORDS

- Zoning land use settlement
- carrying-capacity decisions
- Mechanism chain
- Qualified use
- DGRE under DRDO
- DGRE

**How to use them:** Frame the answer through Zoning land use settlement; define carrying-capacity decisions, connect Mechanism chain with Qualified use to explain the mechanism, and use DGRE under DRDO for the decisive comparison or qualification.

## VISUAL FIRST

ZONING LAND USE SETTLEMENT AND CARRYING-CAPACITY DECISIONS

01. DGRE role

STATUS / CATEGORY FIREWALL -> Do not infer safe outcomes from a map, bulletin, sensor, scheme or structure.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.

## SIMPLE SYSTEM EXAMPLE

Read Zoning land use settlement and carrying-capacity decisions as a sequence: identify DGRE role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not infer safe outcomes from a map, bulletin, sensor, scheme or structure. This keeps DGRE role in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.

## EXAMINER CAUTION

- Do not infer safe outcomes from a map, bulletin, sensor, scheme or structure.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Make zonation alter siting construction and route choices.

## MINI RECAP

- **Mechanism chain:** DGRE role
- **Qualified use:** Make zonation alter siting construction and route choices.

## CLOSING RECALL FLOW - CORE - ZONING LAND USE SETTLEMENT AND CARRYING-CAPACITY DECISIONS

START / CONCEPT: CORE - Zoning land use settlement and carrying-capacity decisions

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EXACT TERMS: Zoning land use settlement · carrying-capacity decisions · Mechanism chain · Qualified use · DGRE under DRDO · DGRE

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MECHANISM / ARGUMENT: Mechanism chain: DGRE role Qualified use: Make zonation alter siting construction and route choices.

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v

CONSEQUENCE / CONTRAST: DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.

|

v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not infer safe outcomes from a map, bulletin, sensor, scheme or structure.

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ANSWER-GRABBING FORMULATION: Read Zoning land use settlement and carrying-capacity decisions as a sequence: identify DGRE role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## SESSION 11 - CORE - DRAINAGE STABILISATION BIOENGINEERING AND ENGINEERING LIMITS

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Drainage stabilisation bioengineering and engineering limits as a sequence: identify NRSC-ISRO role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: NRSC-ISRO role - Glacial-lake monitoring Qualified use: Evaluate site-specific works, maintenance, residual risk and avoidance.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Drainage stabilisation bioengineering and engineering limits as a sequence: identify NRSC-ISRO role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- **Drainage stabilisation bioengineering**
- **engineering limits**
- **Mechanism chain**
- **Qualified use**
- **NRSC**
- **ISRO**

**How to use them:** Frame the answer through Drainage stabilisation bioengineering; define engineering limits, connect Mechanism chain with Qualified use to explain the mechanism, and use NRSC for the decisive comparison or

qualification.

## VISUAL FIRST

DRAINAGE STABILISATION BIOENGINEERING AND ENGINEERING LIMITS

01. NRSC-ISRO role

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02. Glacial-lake monitoring

STATUS / CATEGORY FIREWALL -> Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

## SIMPLE SYSTEM EXAMPLE

Read Drainage stabilisation bioengineering and engineering limits as a sequence: identify NRSC-ISRO role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms. This keeps NRSC-ISRO role and Glacial-lake monitoring in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.
- Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

## EXAMINER CAUTION

- Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Evaluate site-specific works, maintenance, residual risk and avoidance.

## MINI RECAP

- **Mechanism chain:** NRSC-ISRO role -> Glacial-lake monitoring
- **Qualified use:** Evaluate site-specific works, maintenance, residual risk and avoidance.

## CLOSING RECALL FLOW - CORE - DRAINAGE STABILISATION BIOENGINEERING AND ENGINEERING LIMITS

START / CONCEPT: CORE - Drainage stabilisation bioengineering and engineering limits

|  
v

EXACT TERMS: Drainage stabilisation bioengineering · engineering limits · Mechanism chain · Qualified use · NRSC · ISRO

|  
v

MECHANISM / ARGUMENT: Mechanism chain: NRSC-ISRO role - Glacial-lake monitoring Qualified use: Evaluate site-specific works, maintenance, residual risk and avoidance.

|  
v

CONSEQUENCE / CONTRAST: NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.

|  
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms.

|  
v

ANSWER-GRABBING FORMULATION: Read Drainage stabilisation bioengineering and engineering limits as a sequence: identify NRSC-ISRO role, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## SESSION 12 - CORE - ROAD ROUTE HYDROPOWER AND CRITICAL-SERVICE PLANNING

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Road route hydropower and critical-service planning as a sequence: identify Zoning and land use, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Road route hydropower and critical-service planning as a sequence: identify Zoning and land use, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- Road route hydropower
- critical-service planning
- Mechanism chain
- Qualified use
- Susceptibility
- Read Road

**How to use them:** Frame the answer through Road route hydropower; define critical-service planning, connect Mechanism chain with Qualified use to explain the mechanism, and use Susceptibility for the decisive comparison or qualification.

## VISUAL FIRST

ROAD ROUTE HYDROPOWER AND CRITICAL-SERVICE PLANNING

01. Zoning and land use

STATUS / CATEGORY FIREWALL -> Do not confuse intense rainfall or a cloudburst with the mass movement it may trigger.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

## SIMPLE SYSTEM EXAMPLE

Read Road route hydropower and critical-service planning as a sequence: identify Zoning and land use, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not confuse intense rainfall or a cloudburst with the mass movement it may trigger. This keeps Zoning and land use in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

## EXAMINER CAUTION

- Do not confuse intense rainfall or a cloudburst with the mass movement it may trigger.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Balance connectivity and development with drainage alternatives and safe operation.

## MINI RECAP

- **Mechanism chain:** Zoning and land use
- **Qualified use:** Balance connectivity and development with drainage alternatives and safe operation.

## CLOSING RECALL FLOW - CORE - ROAD ROUTE HYDROPOWER AND CRITICAL-SERVICE PLANNING

START / CONCEPT: CORE - Road route hydropower and critical-service planning

|

v

EXACT TERMS: Road route hydropower · critical-service planning · Mechanism chain · Qualified use · Susceptibility · Read Road

|

v

MECHANISM / ARGUMENT: Mechanism chain: Zoning and land use Qualified use: Balance connectivity and development with drainage alternatives and safe operation.

|

v

CONSEQUENCE / CONTRAST: Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

|

v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not confuse intense rainfall or a cloudburst with the mass movement it may trigger.

|

v

ANSWER-GRABBING FORMULATION: Read Road route hydropower and critical-service planning as a sequence: identify Zoning and land use, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

# SESSION 13 - CORE SYNTHESIS - WARNING CLOSURE EVACUATION DRILLS AND ACCESSIBLE COMMUNICATION

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Warning closure evacuation drills and accessible communication as a sequence: identify Slope and drainage measures, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Slope and drainage measures Qualified use: Link uncertainty to pre-agreed closure or evacuation action.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Warning closure evacuation drills and accessible communication as a sequence: identify Slope and drainage measures, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## MUST-WRITE KEYWORDS

- CORE SYNTHESIS
- Warning closure evacuation drills
- accessible communication
- Mechanism chain
- Qualified use
- Surface

**How to use them:** Frame the answer through CORE SYNTHESIS; define Warning closure evacuation drills, connect accessible communication with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

## VISUAL FIRST

WARNING CLOSURE EVACUATION DRILLS AND ACCESSIBLE COMMUNICATION

01. Slope and drainage measures

STATUS / CATEGORY FIREWALL -> Do not equate susceptibility, hazard and risk maps.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope.

## SIMPLE SYSTEM EXAMPLE

Read Warning closure evacuation drills and accessible communication as a sequence: identify Slope and drainage measures, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not equate susceptibility, hazard and risk maps. This keeps Slope and drainage measures in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope.

## EXAMINER CAUTION

- Do not equate susceptibility, hazard and risk maps.

### EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Link uncertainty to pre-agreed closure or evacuation action.

### MINI RECAP

- **Mechanism chain:** Slope and drainage measures
- **Qualified use:** Link uncertainty to pre-agreed closure or evacuation action.

### CLOSING RECALL FLOW - CORE SYNTHESIS - WARNING CLOSURE EVACUATION DRILLS AND ACCESSIBLE COMMUNICATION

START / CONCEPT: CORE SYNTHESIS - Warning closure evacuation drills and accessible communication

|

v

EXACT TERMS: CORE SYNTHESIS · Warning closure evacuation drills · accessible communication · Mechanism chain · Qualified use · Surface

|

v

MECHANISM / ARGUMENT: Mechanism chain: Slope and drainage measures Qualified use: Link uncertainty to pre-agreed closure or evacuation action.

|

v

CONSEQUENCE / CONTRAST: Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope.

|

v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not equate susceptibility, hazard and risk maps.

|

v

ANSWER-GRABBING FORMULATION: Read Warning closure evacuation drills and accessible communication as a sequence: identify Slope and drainage measures, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## SESSION 14 - CORE SYNTHESIS - MULTI-AGENCY MANDATES UNCERTAINTY AND LOCAL KNOWLEDGE

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read Multi-agency mandates uncertainty and local knowledge as a sequence: identify Route and settlement planning, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Route and settlement planning - Warning and evacuation Qualified use: Assign the data owner, warning authority and local action owner.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Multi-agency mandates uncertainty and local knowledge as a sequence: identify Route and settlement planning, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- **CORE SYNTHESIS**
- **Multi-agency mandates uncertainty**
- **local knowledge**
- **Mechanism chain**
- **Qualified use**
- **Effective**

**How to use them:** Frame the answer through CORE SYNTHESIS; define Multi-agency mandates uncertainty, connect local knowledge with Mechanism chain to explain the mechanism, and use Qualified use for the decisive

comparison or qualification.

## VISUAL FIRST

MULTI-AGENCY MANDATES UNCERTAINTY AND LOCAL KNOWLEDGE

01. Route and settlement planning

|

v

02. Warning and evacuation

STATUS / CATEGORY FIREWALL -> Do not assign GSI's landslide role to DGRE or DGRE's avalanche role to ISRO.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

## SIMPLE SYSTEM EXAMPLE

Read Multi-agency mandates uncertainty and local knowledge as a sequence: identify Route and settlement planning, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not assign GSI's landslide role to DGRE or DGRE's avalanche role to ISRO. This keeps Route and settlement planning and Warning and evacuation in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.
- Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

## EXAMINER CAUTION

- Do not assign GSI's landslide role to DGRE or DGRE's avalanche role to ISRO.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** Assign the data owner, warning authority and local action owner.

## MINI RECAP

- **Mechanism chain:** Route and settlement planning -> Warning and evacuation
- **Qualified use:** Assign the data owner, warning authority and local action owner.

## CLOSING RECALL FLOW - CORE SYNTHESIS - MULTI-AGENCY MANDATES UNCERTAINTY AND LOCAL KNOWLEDGE

START / CONCEPT: CORE SYNTHESIS - Multi-agency mandates uncertainty and local knowledge

|  
v

EXACT TERMS: CORE SYNTHESIS · Multi-agency mandates uncertainty · local knowledge · Mechanism chain  
· Qualified use · Effective

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Route and settlement planning - Warning and evacuation  
Qualified use: Assign the data owner, warning authority and local action owner.

|  
v

CONSEQUENCE / CONTRAST: The decisive boundary is Do not assign GSI's landslide role to DGRE or  
DGRE's avalanche role to ISRO.

|  
v

UPSC TRAP / ANSWER-USE: Do not assign GSI's landslide role to DGRE or DGRE's avalanche role to ISRO.

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v

ANSWER-GRABBING FORMULATION: Read Multi-agency mandates uncertainty and local knowledge as a  
sequence: identify Route and settlement planning, follow the named mechanism to its institutional  
response, and stop at the last verified status rather than assuming the next stage.

## SESSION 15 - CORE SYNTHESIS - PYQ SYNTHESIS TECHNOLOGY AND OUTCOME FIREWALL

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Read PYQ synthesis technology and outcome firewall as a sequence: identify Institutional coordination, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

**Technical definition:** Mechanism chain: Institutional coordination - Technology-outcome firewall Qualified use: End with verified receipt evacuation access and loss evidence.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read PYQ synthesis technology and outcome firewall as a sequence: identify Institutional coordination, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

### MUST-WRITE KEYWORDS

- CORE SYNTHESIS
- PYQ synthesis technology
- outcome firewall
- Mechanism chain
- Qualified use
- Subject

**How to use them:** Frame the answer through CORE SYNTHESIS; define PYQ synthesis technology, connect outcome firewall with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

## VISUAL FIRST

PYQ SYNTHESIS TECHNOLOGY AND OUTCOME FIREWALL

01. Institutional coordination

|  
v

02. Technology-outcome firewall

STATUS / CATEGORY FIREWALL -> Do not convert experimental forecasting into nationwide operational warning.

*The rail fixes the institutional sequence and claim boundary before explanation.*

## CORE EXPLANATION

GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

## SIMPLE SYSTEM EXAMPLE

Read PYQ synthesis technology and outcome firewall as a sequence: identify Institutional coordination, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## TECHNICAL DISTINCTION

The decisive boundary is Do not convert experimental forecasting into nationwide operational warning. This keeps Institutional coordination and Technology-outcome firewall in the correct risk category, institutional mandate and evidence status.

## NAMED EVIDENCE AND MECHANISM

- GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.
- A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

## EXAMINER CAUTION

- Do not convert experimental forecasting into nationwide operational warning.

## EXAM LINK

- **Prelims:** Keep hazard, forecast, warning, institution, law, plan, force, fund, scheme and outcome in their exact category.
- **Mains:** End with verified receipt evacuation access and loss evidence.

## MINI RECAP

- **Mechanism chain:** Institutional coordination -> Technology-outcome firewall
- **Qualified use:** End with verified receipt evacuation access and loss evidence.

## COMPLETE BASIC OWNER EVIDENCE BANK

**Subject:** Disaster Management | **Tier:** Must-Do (foundation) | **GS Paper:** GS-III. **Core area:** Landslide/cloudburst hazard management; vulnerability zonation; land-use control and evacuation; Glacial Lake Outburst Flood (GLOF) mechanisms and mitigation. **Grounded in:** *VisionIAS Value Added Material Disaster Management*, PDF pp. 29-35 (landslides/cloudbursts), pp. 78-79 (GLOF model answer); 00\_Master-Framework.md Section 5. **FACT:** = source-grounded | **ANALYSIS:** = analytical inference | **CURRENT:** = current anchor | **WRONG:** = boundary/trap. *Companion:* advanced/10\_Landslides-Avalanches-and-GLOF-Risk.md.

## 1. Visual foundation

SLOPE INSTABILITY (natural + anthropogenic causes)

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LANDSLIDE VULNERABILITY ZONATION

Very High -> High -> Moderate/Low -> Safe areas

|

v

HAZARD MAPPING + LAND-USE RESTRICTION + AFORESTATION

|

v

CASCADING RISK: cloudburst -> flash flood -> landslide

snowpack/slope/weather -> avalanche (DGRE warning)

glacial retreat -> GLOF -> downstream flooding

**Core proposition:** FACT: VisionIAS defines a landslide as "the movement of a mass of rock, debris, or earth down a slope" - a form of "mass wasting" - affecting "at least 15 per cent of the land area" of India, driven fundamentally by the Indian plate's continuous northward movement (~5 cm/year) against the Eurasian plate rendering Himalayan rock "friable, weak and prone to landslides and earthquakes" (PDF p. 31). ANALYSIS: An avalanche - the rapid down-slope movement of a mass of snow and ice - is a mechanistically distinct Himalayan mass-movement hazard, governed by snowpack, slope and weather conditions rather than rock/ debris movement, and is monitored/warned by the Defence Geoinformatics Research Establishment (DGRE), not treated as a sub-type of landslide.

## 2. Essential definitions

| Concept                                         | Exam-ready meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FACT: <b>Landslide</b>                          | Down-slope movement of a mass of rock, debris or earth under gravity - a form of mass wasting (PDF p. 31).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| FACT: <b>Cloudburst</b>                         | Extremely fast, localised heavy rainfall - around 10 cm or more per hour over roughly 20-30 sq km - caused by saturated clouds unable to release rain due to strong upward warm-air currents, until raindrops become too heavy and fall together suddenly (PDF p. 33).                                                                                                                                                                                                                                                                                                                                                |
| FACT: <b>Landslide vulnerability zones</b>      | Very High (young Himalayan/Andaman-Nicobar mountains, high-rainfall steep-slope Western Ghats/Nilgiris/North-East, frequent-earthquake zones, intense human-activity areas); High (similar factors, lower intensity/frequency, excluding Assam plains); Moderate/Low (Trans-Himalayan Ladakh/Spiti, Aravali, rain-shadow Western/Eastern Ghats, Deccan plateau; also mining/subsidence-linked landslides in Jharkhand, Odisha, Chhattisgarh, MP, Maharashtra, AP, Karnataka, TN, Goa, Kerala); Other/safe areas (Rajasthan, Haryana, UP, Bihar, most of West Bengal/Assam, southern coastal regions) (PDF pp. 31-32). |
| ANALYSIS: <b>Avalanche</b>                      | A rapid down-slope movement of a mass of snow, ice and entrained debris, released when gravitational stress on an unstable snowpack exceeds its internal strength - triggered by a combination of snowpack structure (weak layers), slope angle (typically 30-45°), and weather (fresh snowfall load, temperature fluctuation, wind-loading); mechanistically distinct from a landslide (rock/debris, not primarily snow/ice) and from a GLOF (a lake-outburst flood, not a slope-failure mass movement), though all three are Himalayan mass-movement/cryosphere hazards.                                            |
| FACT: <b>Glacial Lake Outburst Flood (GLOF)</b> | Sudden release of a large volume of water from a glacial lake, caused by the failure or overtopping of the natural dam impounding it - whether an ice dam, moraine dam, bedrock dam or other natural dam - releasing melt water, moraines and the lake's contents and causing catastrophic downstream flooding with major geomorphic/socio-economic impact (PDF pp. 26, 78).                                                                                                                                                                                                                                          |

## 3. Risk/problem mechanism

- Landslide natural causes:** heavy rain (the main cause); earthquakes and volcanic explosions destabilising Himalayan rock (PDF p. 32).
- Landslide anthropogenic causes:** deforestation removing soil-binding vegetation; road construction dislodging rock structure and altering slope angles; shifting agriculture in the North-East; and house/hotel construction generating destabilising debris (PDF p. 32).
- Cloudburst mechanism and India-specific pattern:** hilly terrain promotes vertical warm-air currents that trigger cloudbursts most often when Bay-of-Bengal monsoon low-pressure clouds travel northward across the Ganga plains onto the Himalayas; a single disaster event (e.g. Uttarakhand 2013) can involve multiple cloudburst instances (PDF p. 33).
- Avalanche triggers:** ANALYSIS: a weak layer within an accumulated snowpack fails under added stress - fresh/wind-loaded snow, fluctuating temperature (freeze-thaw weakening bonds), rain-on-snow, or a steep slope (most avalanche-prone in the 30-45° range) - a distinct trigger-mechanism from both

landslide (rock/soil/rain-driven) and GLOF (glacial-lake dam failure); concentrated in the high-Himalayan/Ladakh/Siachen snow-bound belt.

5. **GLOF triggers:** glacial retreat from global warming increasing the number/size of glacial lakes; rapid slope movement into the lake and ice-dam melting; anthropogenic radiative-balance alteration (mass tourism, roads, hydropower projects accelerating glacial melt); Black Carbon surface-heating impact; earthquakes; blocked subsurface outflow tunnels; and long-term dam degradation (PDF pp. 78-79).

#### 4. Disaster-management cycle application

- **FACT: Landslide mitigation (NDMA guidelines):** hazard-zone identification with specific-slide stabilisation, monitoring and early-warning systems at selected sites; area-specific measures; hazard mapping; restrictions on construction/roads/dams and large settlements in high-vulnerability zones; large-scale afforestation and water-flow-reducing bund construction; terrace farming replacing Jhumming in the North-East; retaining walls; excavation/construction/ grading codes; landslide insurance/compensation arrangements (PDF pp. 32-33).
- **FACT: Cloudburst mitigation:** NDMA has not issued cloudburst-specific guidelines - flood and landslide guidelines apply by extension; measures include settlement on hard rock/firm ground rather than valleys, vigilance for slope changes, banning indiscriminate construction in landslide-affected areas, surface/subsurface drainage planning, bioengineering (local vegetation plus engineering), and dense rain-gauge networks in vulnerable areas (PDF pp. 33-34).
- **ANALYSIS: Avalanche mitigation:** DGRE (Defence Geoinformatics Research Establishment, DRDO) issues avalanche warnings/forecasts for the Himalayan/Ladakh region using snowpack, slope and weather data; complementary measures include avalanche zoning of settlements/roads, defence/civilian route-closure protocols during high-risk periods, and protective structures (snow fences, deflection walls) - a distinct warning-and-mitigation chain from both landslide guidelines and GLOF monitoring.
- **FACT: GLOF mitigation:** early-warning access with continuous remote-sensing monitoring; dam remediation (artificial dams, tunnels, open cuts, concrete outflows, flood-protection walls); GLOF hazard mapping; identification of safe evacuation sites and vulnerable communities; community-based early-warning systems (PDF p. 79).

#### 5. Institutions and policy tools

- **FACT: ISRO** - among the organisations engaged in monitoring glacial lakes and other water bodies in the Himalayan region's Indian river basins (PDF p. 79).
- **CURRENT: Geological Survey of India (GSI)** - the nodal agency for landslide hazard. GSI completed **National Landslide Susceptibility Mapping at 1:50,000 over about 4.3 lakh sq km** of landslide-prone terrain in parts of **19 States/UTs**, classifying terrain as low, moderate or high susceptibility; maps and the landslide inventory are served through the **Bhukosh** and NGDR portals (PIB/Ministry of Mines, 20 August 2025).
- **CURRENT: National Landslide Forecasting Centre (NLFC)** - inaugurated **19 July 2024**, with regional landslide forecasts beginning **20 July 2024** for **Darjeeling, Kalimpong and the Nilgiris**. By December 2025 the Regional Landslide Forecasting System covered **21 districts - 4 operational and 17 experimental** - issuing daily next-48-hour forecasts at taluk/sub-divisional scale, disseminated via the **Bhusanket** web portal and **Bhooskhalan** mobile app (PIB/MoES, 17 December 2025). **ANALYSIS:** Distinguish the **operational** from the **experimental** districts - the second set is not a service guarantee.
- **CURRENT: \*\*NDMA's National Landslide Risk Management Strategy (September 2019) is the governing national strategy document; NDMA's Guidelines on Management of Landslides and Snow Avalanches (June 2009)\*\*** remain the underlying guideline.
- **CURRENT: \*\*NDMA's Guidelines on Management of Glacial Lake Outburst Floods**

(October 2020) **and the National GLOF Risk Mitigation Programme (NGRMP) - outlay Rs 150 crore (Rs 135 crore from the National Disaster Mitigation Fund plus Rs 15 crore State contribution), covering Arunachal Pradesh, Himachal Pradesh, Sikkim and Uttarakhand (MHA/ PIB, 4 February 2025). The official dynamic list contained 195 high-risk glacial lakes\*\* as of 4 December 2024.**

- ANALYSIS: **DGRE (Defence Geoinformatics Research Establishment, DRDO) -**

India's dedicated avalanche-warning institution, issuing snow/ avalanche forecasts and warnings for the Himalayan/Ladakh region based on snowpack, slope and weather monitoring; the correct institutional anchor for any avalanche-warning-specific claim, distinct from GSI's landslide-forecasting role and ISRO's glacial-lake monitoring role.

- FACT: **Copenhagen's cloudburst master plan** - cited as a useful

international model, aiming to decouple 30-40% of excess stormwater from the combined sewer system through concretisation, canal creation and greening (PDF p. 34).

## 6. India applications and examples

- FACT: VisionIAS names five worst Indian cloudburst events: the Great Musi

flood of 1908 near Hyderabad; Mumbai Cloudburst 2005; Jammu and Kashmir cloudburst 2010; and the Uttarakhand cloudbursts of 1998 and 2013 (PDF p. 34).

- ANALYSIS: **PYQ mapping:** the 2025 mining-environmental-hazard question

("Why is mining considered an environmental hazard?") is **adjacent-only** - cross-linked here specifically for slope/ subsidence risk (mining-linked landslides named in the Moderate/Low Vulnerability Zone, Section 2), not absorbed as a direct DM question.

## 7. Must-Know Facts for Prelims

- FACT: A cloudburst is classified at roughly 10 cm or more rainfall per

hour over ~20-30 sq km.

- FACT: Landslides affect at least 15% of India's land area (document-period

estimate).

- FACT: Very High vulnerability landslide zones include the Himalayas,

Andaman and Nicobar, Western Ghats/Nilgiris and North-East India.

- FACT: Mining- and subsidence-linked landslides are concentrated in

Jharkhand, Odisha, Chhattisgarh, MP, Maharashtra, AP, Karnataka, TN, Goa and Kerala.

- FACT: GLOFs result from the failure or overtopping of a glacial lake's

natural dam - ice, moraine, bedrock or other natural dam - releasing melt water, moraines and the lake's contents.

- CURRENT: GSI completed **\*\*National Landslide Susceptibility Mapping** at

1:50,000 over ~4.3 lakh sq km **across parts of 19 States/UTs; outputs are served through Bhukosh\*\*.**

- CURRENT: The **National Landslide Forecasting Centre** was inaugurated

**19 July 2024**, forecasts beginning 20 July 2024 for **Darjeeling, Kalimpong and the Nilgiris**; by December 2025 the Regional Landslide Forecasting System covered **21 districts (4 operational, 17 experimental)** with daily 48-hour forecasts, disseminated via **Bhusanket** (portal) and **Bhooskhalan** (app).

- CURRENT: NDMA's **National Landslide Risk Management Strategy** dates to

**September 2019**; its **GLOF Guidelines** to **October 2020**.

- CURRENT: The **National GLOF Risk Mitigation Programme** has an outlay of

**Rs 150 crore (Rs 135 crore NDMF + Rs 15 crore State)** covering **Arunachal Pradesh, Himachal Pradesh, Sikkim and Uttarakhand**; the official list carried **195 high-risk glacial lakes** as of 4 December 2024.

- ANALYSIS: An avalanche is triggered by snowpack instability, slope angle

(typically 30-45°) and weather (snow load, temperature, wind); DGRE (DRDO) is India's dedicated avalanche-warning institution for the Himalayan/Ladakh region.

## 8. WRONG: Traps and stale-claim cautions

- WRONG: Landslide, cloudburst, avalanche and GLOF are the same hazard. ->

Each has a distinct mechanism: landslide (gravity-driven mass movement), cloudburst (extreme localised rainfall event), GLOF (sudden release from a glacial lake caused by failure/overtopping of its natural dam - ice, moraine, bedrock or other), avalanche (rapid down-slope movement of a snow/ice mass triggered by snowpack, slope and weather conditions - Section 2) - related but mechanistically distinct hazards.

- WRONG: NDMA has issued dedicated national cloudburst guidelines. -> The source explicitly states NDMA "has not released guidelines specifically on Cloudbursts"; flood and landslide guidelines apply by extension (PDF p. 33).

- WRONG: Cloudbursts can be reliably predicted with current technology. ->

VisionIAS notes it is "very difficult to predict the cloud bursts due to its very small scale in space and time," even though IMD provides real-time rainfall situation/intensity and forecasts (PDF p. 34-35).

- WRONG: The "15% of land area" and vulnerability-zone lists are current without verification. -> Treat as document-period estimates; verify against current GSI/NDMA sources.

## 9. CURRENT: Current official anchor

- CURRENT: \*\*GSI's National Landslide Forecasting Centre and Bhooskhalan platform\*\*, checked as of the 18 July 2026 research cutoff, are the correct current anchors for landslide-forecasting/monitoring service status and current vulnerability-zone data - VisionIAS's zonation table and "15% of land area" estimate must be cross-checked against this source for any numerically precise claim.

## 11. Mains framework

1. **Name the specific hazard precisely** (landslide, cloudburst, avalanche, GLOF) rather than treating them as interchangeable (Section 8).
2. **Cite the specific vulnerability zone** relevant to the region asked about (Section 2), not a generic "the Himalayas are landslide-prone" statement.
3. **Distinguish natural from anthropogenic causes** (Section 3), naming specific anthropogenic drivers (mining, road construction, shifting agriculture) precisely.
4. **State honestly that cloudburst prediction remains very difficult** (Section 8) even with IMD's real-time monitoring capability.
5. \*\*Close by linking landslide/GLOF risk to Himalayan development-risk tension\*\* (topic 01's risk-creation loop) and multi-hazard cascading risk (topic 04).

## 12. Study links

- FACT: Advanced companion:  
advanced/10\_Landslides-Avalanches-and-GLOF-Risk.md.
- FACT: 00\_Master-Framework.md Section 5 - landslide/GLOF is classified under the geological hazard family alongside topics 05 and 06.
- ANALYSIS: **Downstream topics in this folder:** Topic 05 develops earthquake-linked seismic risk feeding into landslide triggers; topic 04 develops multi-hazard early-warning generally; topic 15 develops the climate-linked glacial-retreat/adaptation dimension.

## 13. Core-only answer architecture - mapped, cascading Himalayan risk

**Core firewall:** the 2019/2021 landslide routes terminate here. "Himalayas are fragile" alone is not an answer: Core supplies a comparison, a zonation mechanism, a named strategy and a qualified development verdict.

### 13.1 Claim-to-evidence bank

| Claim                                                                             | Named evidence/example                                                                                                                                                                       | Significance                                                                              | Limitation/qualification                                                                                                 |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Landslide risk is a hazard-slope-development interaction.                         | Young/friable, tectonically active Himalaya; intense rain, earthquakes, road cutting, deforestation, shifting cultivation and slope loading.                                                 | Converts physical description into an exposure/vulnerability argument.                    | Do not imply every Himalayan road or rainfall event causes a slide; use site-specific assessment.                        |
| Himalayan and Western-Ghat cases share rainfall/slope risk but differ in setting. | Himalaya: young, seismic, high-relief terrain and development pressure; Western Ghats/Nilgiris: steep weathered slopes under intense monsoon rain, compounded by road cutting/deforestation. | Directly answers the 2021 "differentiate" directive rather than listing one common cause. | Physical-process depth belongs to Geography; this file owns the risk-management comparison.                              |
| Zonation changes decisions before a failure.                                      | GSI susceptibility mapping, NDMA National Landslide Risk Management Strategy (2019), construction/road/settlement restrictions in high-vulnerability zones.                                  | A map becomes a land-use, monitoring and evacuation decision tool.                        | A susceptibility map is not a site-specific forecast or a substitute for enforcement.                                    |
| Himalayan mass-movement hazards require separate warning/mitigation chains.       | Landslide/cloudburst; DGRE avalanche warnings; GLOF lake/dam-failure mechanism, NDMA GLOF guidelines and NGRMP.                                                                              | Prevents the common landslide-avalanche-GLOF conflation.                                  | Operational versus experimental forecast coverage and dynamic lake lists must be dated; do not claim nationwide warning. |

### 13.2 Executable spines

- **10 marks - 2021 Himalaya versus Western Ghats:** write a two-column

comparison (geology/relief and seismicity; rainfall/weathering; human interventions; response tool), then one shared conclusion: risk-sensitive siting, drainage and slope management are needed in both but cannot be one-size-fits-all.

- **15 marks - 2019/2021 zonation/NLRMS route:** thesis that mapping is

a preventive governance instrument. Explain inputs (slope, geology, rainfall, seismicity, land use), translate very-high/high zones into no/conditional construction, monitoring, drainage and evacuation, name GSI/NLRMS and include the enforcement/settlement trade-off.

- **20 marks - Himalayan development and cascading risk:** use

earthquake -> landslide -> blockage/outburst, cloudburst -> flash flood -> landslide, and glacial retreat -> GLOF chains. Balance roads, tourism and hydropower livelihoods against carrying capacity, risk screening, warnings and downstream evacuation; conclude with risk-informed rather than blanket anti-development planning.

## CLOSING RECALL FLOW - CORE SYNTHESIS - PYQ SYNTHESIS TECHNOLOGY AND OUTCOME FIREWALL

START / CONCEPT: CORE SYNTHESIS - PYQ synthesis technology and outcome firewall

|  
v

EXACT TERMS: CORE SYNTHESIS · PYQ synthesis technology · outcome firewall · Mechanism chain · Qualified use · Subject

|  
v

MECHANISM / ARGUMENT: Mechanism chain: Institutional coordination - Technology-outcome firewall  
Qualified use: End with verified receipt evacuation access and loss evidence.

|  
v

CONSEQUENCE / CONTRAST: The decisive boundary is Do not convert experimental forecasting into nationwide operational warning.

|  
v

UPSC TRAP / ANSWER-USE: Cite the specific vulnerability zone relevant to the region asked about (Section 2), not a generic "the Himalayas are landslide-prone" statement.

|  
v

ANSWER-GRABBING FORMULATION: Read PYQ synthesis technology and outcome firewall as a sequence: identify Institutional coordination, follow the named mechanism to its institutional response, and stop at the last verified status rather than assuming the next stage.

## BASIC MCQS / REMEDIATION

### Q1. Which statement correctly identifies Landslide?

A. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. B. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. C. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. D. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

**Answer: A. Explanation: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q2. Which option preserves the risk or institutional boundary of Landslide?

A. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. B. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. C. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. D. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

**Answer: B. Explanation: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q3. Which statement uses Landslide without changing its hazard, mandate or status?

A. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. D. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

**Answer: C. Explanation: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q4. Which option avoids the standard UPSC close-option trap about Landslide?

A. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. B. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. C. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. D. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

**Answer: D. Explanation: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q5. Which statement correctly identifies Rockfall?

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

**Answer: A. Explanation: Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q6. Which option preserves the risk or institutional boundary of Rockfall?

A. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. B. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. C. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. D. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

**Answer: B. Explanation: Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q7. Which statement uses Rockfall without changing its hazard, mandate or status?

A. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. D. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

**Answer: C. Explanation: Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q8. Which option avoids the standard UPSC close-option trap about Rockfall?

A. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. B. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow.

**Answer: D. Explanation: Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q9. Which statement correctly identifies Debris flow?

A. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.

**Answer: A. Explanation: A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q10. Which option preserves the risk or institutional boundary of Debris flow?

A. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. B. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

**Answer: B. Explanation: A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q11. Which statement uses Debris flow without changing its hazard, mandate or status?**

A. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

**Answer: C. Explanation: A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q12. Which option avoids the standard UPSC close-option trap about Debris flow?**

A. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. B. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

**Answer: D. Explanation: A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q13. Which statement correctly identifies Avalanche?**

A. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.

**Answer: A. Explanation: An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q14. Which option preserves the risk or institutional boundary of Avalanche?**

A. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. B. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. C. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. D. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.

**Answer: B. Explanation: An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q15. Which statement uses Avalanche without changing its hazard, mandate or status?**

A. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. B. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. C. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

**Answer: C. Explanation: An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q16. Which option avoids the standard UPSC close-option trap about Avalanche?**

A. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. B. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. C. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. D. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.

**Answer: D. Explanation: An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q17. Which statement correctly identifies GLOF?**

A. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. B. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

**Answer: A. Explanation: A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q18. Which option preserves the risk or institutional boundary of GLOF?

A. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. B. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. C. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

**Answer: B. Explanation: A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q19. Which statement uses GLOF without changing its hazard, mandate or status?

A. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. B. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

**Answer: C. Explanation: A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q20. Which option avoids the standard UPSC close-option trap about GLOF?

A. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. B. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. C. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. D. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

**Answer: D. Explanation: A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q21. Which statement correctly identifies Susceptibility versus hazard?

A. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. B. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

**Answer: A. Explanation:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

**Q22. Which option preserves the risk or institutional boundary of Susceptibility versus hazard?**

A. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. D. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.

**Answer: B. Explanation:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

**Q23. Which statement uses Susceptibility versus hazard without changing its hazard, mandate or status?**

A. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. B. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. C. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

**Answer: C. Explanation:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

**Q24. Which option avoids the standard UPSC close-option trap about Susceptibility versus hazard?**

A. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. B. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. C. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. D. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

**Answer: D. Explanation:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

### Q25. Which statement correctly identifies Risk?

A. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. B. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. C. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. D. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.

**Answer: A. Explanation: Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q26. Which option preserves the risk or institutional boundary of Risk?

A. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. B. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. C. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

**Answer: B. Explanation: Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q27. Which statement uses Risk without changing its hazard, mandate or status?

A. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. B. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

**Answer: C. Explanation: Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q28. Which option avoids the standard UPSC close-option trap about Risk?

A. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. B. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. C. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. D. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.

**Answer: D. Explanation: Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q29. Which statement correctly identifies Conditioning factors?**

A. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. B. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. C. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. D. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.

**Answer: A. Explanation: Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q30. Which option preserves the risk or institutional boundary of Conditioning factors?**

A. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. B. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. C. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. D. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.

**Answer: B. Explanation: Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q31. Which statement uses Conditioning factors without changing its hazard, mandate or status?**

A. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. B. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. C. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

**Answer: C. Explanation: Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q32. Which option avoids the standard UPSC close-option trap about Conditioning factors?

A. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. B. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. C. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

**Answer: D. Explanation: Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q33. Which statement correctly identifies Triggers?

A. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. B. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. C. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. D. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.

**Answer: A. Explanation: Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q34. Which option preserves the risk or institutional boundary of Triggers?

A. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. B. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. C. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. D. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.

**Answer: B. Explanation: Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q35. Which statement uses Triggers without changing its hazard, mandate or status?

A. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. B. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. C. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. D. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.

**Answer: C. Explanation: Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q36. Which option avoids the standard UPSC close-option trap about Triggers?**

A. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. B. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. C. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. D. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.

**Answer: D. Explanation: Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q37. Which statement correctly identifies Cascading risk?**

A. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. B. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. C. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

**Answer: A. Explanation: Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q38. Which option preserves the risk or institutional boundary of Cascading risk?**

A. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. B. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. C. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. D. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.

**Answer: B. Explanation: Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q39. Which statement uses Cascading risk without changing its hazard, mandate or status?**

A. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. D. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

**Answer: C. Explanation: Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q40. Which option avoids the standard UPSC close-option trap about Cascading risk?**

A. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.

**Answer: D. Explanation: Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q41. Which statement correctly identifies GSI role?**

A. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. B. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. C. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. D. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.

**Answer: A. Explanation: GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q42. Which option preserves the risk or institutional boundary of GSI role?**

A. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. B. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. C. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. D. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

**Answer: B. Explanation: GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q43. Which statement uses GSI role without changing its hazard, mandate or status?**

A. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. D. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

**Answer: C. Explanation: GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q44. Which option avoids the standard UPSC close-option trap about GSI role?**

A. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. B. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. C. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

**Answer: D. Explanation: GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q45. Which statement correctly identifies DGRE role?**

A. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. B. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. C. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. D. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

**Answer: A. Explanation: DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

#### Q46. Which option preserves the risk or institutional boundary of DGRE role?

A. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. B. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. C. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. D. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope.

**Answer: B. Explanation: DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

#### Q47. Which statement uses DGRE role without changing its hazard, mandate or status?

A. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. D. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

**Answer: C. Explanation: DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

#### Q48. Which option avoids the standard UPSC close-option trap about DGRE role?

A. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. B. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. C. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. D. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.

**Answer: D. Explanation: DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

#### Q49. Which statement correctly identifies NRSC-ISRO role?

A. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. B. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

**Answer: A. Explanation: NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q50. Which option preserves the risk or institutional boundary of NRSC-ISRO role?**

A. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. B. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

**Answer: B. Explanation: NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q51. Which statement uses NRSC-ISRO role without changing its hazard, mandate or status?**

A. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. B. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. C. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. D. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

**Answer: C. Explanation: NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q52. Which option avoids the standard UPSC close-option trap about NRSC-ISRO role?**

A. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. B. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. C. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. D. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.

**Answer: D. Explanation: NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q53. Which statement correctly identifies Glacial-lake monitoring?

A. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

**Answer: A. Explanation: Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q54. Which option preserves the risk or institutional boundary of Glacial-lake monitoring?

A. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. B. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

**Answer: B. Explanation: Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q55. Which statement uses Glacial-lake monitoring without changing its hazard, mandate or status?

A. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. B. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. C. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. D. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

**Answer: C. Explanation: Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q56. Which option avoids the standard UPSC close-option trap about Glacial-lake monitoring?

A. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. B. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

**Answer: D. Explanation: Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q57. Which statement correctly identifies Zoning and land use?**

A. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. B. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

**Answer: A. Explanation: Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q58. Which option preserves the risk or institutional boundary of Zoning and land use?**

A. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. D. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

**Answer: B. Explanation: Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q59. Which statement uses Zoning and land use without changing its hazard, mandate or status?**

A. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. B. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. C. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. D. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

**Answer: C. Explanation: Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q60. Which option avoids the standard UPSC close-option trap about Zoning and land use?

A. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. B. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. C. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. D. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

**Answer: D. Explanation: Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q61. Which statement correctly identifies Slope and drainage measures?

A. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. B. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. C. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. D. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

**Answer: A. Explanation: Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q62. Which option preserves the risk or institutional boundary of Slope and drainage measures?

A. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. B. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. C. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. D. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

**Answer: B. Explanation: Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q63. Which statement uses Slope and drainage measures without changing its hazard, mandate or status?

A. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. B. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

**Answer: C. Explanation: Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q64. Which option avoids the standard UPSC close-option trap about Slope and drainage measures?**

A. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. B. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope.

**Answer: D. Explanation: Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q65. Which statement correctly identifies Route and settlement planning?**

A. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. B. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. C. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. D. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

**Answer: A. Explanation: Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q66. Which option preserves the risk or institutional boundary of Route and settlement planning?**

A. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. B. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

**Answer: B. Explanation: Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q67. Which statement uses Route and settlement planning without changing its hazard, mandate or status?**

A. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. B. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. C. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. D. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow.

**Answer: C. Explanation: Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q68. Which option avoids the standard UPSC close-option trap about Route and settlement planning?**

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. C. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. D. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

**Answer: D. Explanation: Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q69. Which statement correctly identifies Warning and evacuation?**

A. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. B. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

**Answer: A. Explanation: Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q70. Which option preserves the risk or institutional boundary of Warning and evacuation?**

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

**Answer: B. Explanation: Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q71. Which statement uses Warning and evacuation without changing its hazard, mandate or status?**

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. C. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. D. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

**Answer: C. Explanation: Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q72. Which option avoids the standard UPSC close-option trap about Warning and evacuation?**

A. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. B. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. C. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. D. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

**Answer: D. Explanation: Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q73. Which statement correctly identifies Institutional coordination?**

A. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. B. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. C. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. D. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

**Answer: A. Explanation: GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q74. Which option preserves the risk or institutional boundary of Institutional coordination?**

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. C. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. D. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

**Answer: B. Explanation: GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q75. Which statement uses Institutional coordination without changing its hazard, mandate or status?**

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. C. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. D. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

**Answer: C. Explanation: GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q76. Which option avoids the standard UPSC close-option trap about Institutional coordination?**

A. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. B. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

**Answer: D. Explanation: GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

**Q77. Which statement correctly identifies Technology-outcome firewall?**

A. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. B. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. C. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. D. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

**Answer: A. Explanation: A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q78. Which option preserves the risk or institutional boundary of Technology-outcome firewall?

A. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. B. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. C. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. D. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.

**Answer: B. Explanation: A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q79. Which statement uses Technology-outcome firewall without changing its hazard, mandate or status?

A. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. B. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

**Answer: C. Explanation: A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

### Q80. Which option avoids the standard UPSC close-option trap about Technology-outcome firewall?

A. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

**Answer: D. Explanation: A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.**

## PYQS AND ANSWER PRACTICE

## AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

All three cards are verified direct routes: 2019 landslide zonation, 2021 Himalayan-versus-Western-Ghats causes, and 2021 landslide causes/effects/National Landslide Risk Management Strategy. Avalanche and GLOF content is not misrepresented as printed in those stems.

### PYQ DEMAND CARD 1 - 2019 GS-III

**Demand:** Explain landslide hazard zonation mapping and its role in preparedness and mitigation.

**Status:** Verified direct routing: Explain · 15 marks · 250 words; the answer distinguishes susceptibility, hazard and risk and converts mapping into decisions.

**Model solution: Landslide:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Susceptibility versus hazard:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. **Risk:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. **Conditioning factors:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **GSI role:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Zoning and land use:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Slope and drainage measures:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. **Warning and evacuation:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Technology-outcome firewall:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

### PYQ DEMAND CARD 2 - 2021 GS-I

**Demand:** Differentiate causes of landslides in the Himalayan region and Western Ghats.

**Status:** Verified direct routing: Differentiate · 10 marks · 150 words; use a comparison of conditioning factors, rainfall, seismicity and intervention without unsupported event statistics.

**Model solution: Landslide:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Conditioning factors:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Triggers:** Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. **Cascading risk:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. **Zoning and land use:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Slope and drainage measures:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

### PYQ DEMAND CARD 3 - 2021 GS-III

**Demand:** Describe landslide causes, effects and the National Landslide Risk Management Strategy.

**Status:** Verified direct routing: Describe · 15 marks · 250 words; strategy use remains bounded to mapping, prevention, warning, capacity and land-use functions.

**Model solution: Landslide:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Risk:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.

**Conditioning factors:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Triggers:** Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. **GSI role:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Zoning and land use:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Slope and drainage measures:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. **Route and settlement planning:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

**Warning and evacuation:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Institutional coordination:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

**Technology-outcome firewall:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

### ORIGINAL MAINS 1 - 10 MARKS

**Question:** Distinguish landslide, rockfall, debris flow, avalanche and GLOF. Answer in about 150 words.

**Model thesis: Claim:** Landslide. **Named evidence/example:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

**Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Rockfall. **Named evidence/example:** Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Debris flow. **Named evidence/example:** A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the

mandate-to-implementation-to-outcome boundary. **Claim:** Avalanche. **Named evidence/example:** An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category,

institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

**Claim -> named evidence -> analysis -> qualification:**

- A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.
- Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow.
- A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.
- An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.
- A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

**Qualified conclusion: Claim:** Landslide. **Named evidence/example:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Rockfall. **Named evidence/example:** Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Debris flow. **Named evidence/example:** A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Avalanche. **Named evidence/example:** An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

## ORIGINAL MAINS 2 - 10 MARKS

**Question:** Differentiate landslide susceptibility, hazard and risk. Answer in about 150 words.

**Model thesis: Claim:** Susceptibility versus hazard. **Named evidence/example:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk. **Named evidence/example:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhuvan material distinguishes mapping and

experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

**Claim -> named evidence -> analysis -> qualification:**

- Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.
- Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.
- GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.
- A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

**Qualified conclusion: Claim:** Susceptibility versus hazard. **Named evidence/example:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk. **Named evidence/example:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

### ORIGINAL MAINS 3 - 15 MARKS

**Question:** Analyse conditioning factors, triggers and cascading mountain risk. Answer in about 250 words.

**Model thesis: Claim:** Conditioning factors. **Named evidence/example:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Triggers. **Named evidence/example:** Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and

the mandate-to-implementation-to-outcome boundary. **Claim:** Cascading risk. **Named evidence/example:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

**Claim -> named evidence -> analysis -> qualification:**

- Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.
- Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.
- Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.
- Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

**Qualified conclusion:** **Claim:** Conditioning factors. **Named evidence/example:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Triggers. **Named evidence/example:** Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cascading risk. **Named evidence/example:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

## ORIGINAL MAINS 4 - 15 MARKS

**Question:** Explain glacial-lake monitoring, GLOF warning and downstream evacuation. Answer in about 250 words.

**Model thesis:** **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** NRSC-ISRO role. **Named evidence/example:** NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and

verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Glacial-lake monitoring. **Named evidence/example:** Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional coordination. **Named evidence/example:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

**Claim -> named evidence -> analysis -> qualification:**

- A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.
- NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.
- Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.
- Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.
- GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.
- A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

**Qualified conclusion:** **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** NRSC-ISRO role. **Named evidence/example:** NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Glacial-lake monitoring. **Named evidence/example:** Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named**

**evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional coordination. **Named evidence/example:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

## ORIGINAL MAINS 5 - 20 MARKS

**Question:** Evaluate zoning, drainage, slope treatment, road alignment and settlement planning for landslide risk. Answer in about 300 words.

**Model thesis:** **Claim:** Susceptibility versus hazard. **Named evidence/example:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk. **Named evidence/example:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Conditioning factors. **Named evidence/example:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and land use. **Named evidence/example:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slope and drainage measures. **Named evidence/example:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal

character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

**Claim -> named evidence -> analysis -> qualification:**

- Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.
- Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.
- Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.
- GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.
- Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.
- Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope.
- Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.
- Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.
- A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

**Qualified conclusion:** **Claim:** Susceptibility versus hazard. **Named evidence/example:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk. **Named evidence/example:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Conditioning factors. **Named evidence/example:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and land use. **Named evidence/example:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slope and drainage measures. **Named evidence/example:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

## ORIGINAL MAINS 6 - 20 MARKS

**Question:** Design a coordinated Himalayan landslide-avalanche-GLOF governance framework under uncertainty. Answer in about 300 words.

**Model thesis:** **Claim:** Landslide. **Named evidence/example:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Avalanche. **Named evidence/example:** An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cascading risk. **Named evidence/example:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category,

institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** DGRE role. **Named evidence/example:** DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** NRSC-ISRO role. **Named evidence/example:** NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Glacial-lake monitoring. **Named evidence/example:** Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and land use. **Named evidence/example:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional coordination. **Named evidence/example:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

**Claim -> named evidence -> analysis -> qualification:**

- A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.
- An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.
- A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.
- Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.

- GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.
- DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.
- NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.
- Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.
- Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.
- Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.
- Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.
- GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.
- A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

**Qualified conclusion: Claim:** Landslide. **Named evidence/example:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Avalanche. **Named evidence/example:** An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cascading risk. **Named evidence/example:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** DGRE role. **Named evidence/example:** DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** NRSC-ISRO role. **Named evidence/example:** NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the

mandate-to-implementation-to-outcome boundary. **Claim:** Glacial-lake monitoring. **Named evidence/example:** Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and land use. **Named evidence/example:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional coordination. **Named evidence/example:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

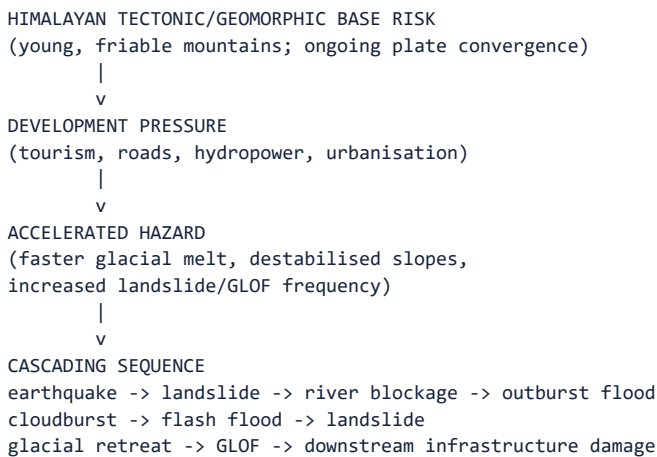
## OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

**Subject:** Disaster Management | **Tier:** Advanced | **GS Paper:** GS-III. **Core area:** Himalayan development-risk analysis; multi-hazard cascades; GLOF governance and project-risk screening. **Grounded in:** *VisionIAS Value Added Material Disaster Management*, PDF pp. 29-35, 78-79; 00\_Master - Framework .md Sections 5, 7-8. **FACT:** = source-grounded | **ANALYSIS:** = inference/analysis | **CURRENT:** = current anchor | **WRONG:** = boundary/trap. **Companion:** basic/10\_Landslides-Avalanches-and-GLOF-Risk.md.

### 1. Advanced proposition

**Thesis:** ANALYSIS: VisionIAS's own GLOF-trigger analysis names "mass tourism" and "developmental interventions such as roads and hydropower projects" as accelerants of glacial melting (PDF p. 79) - meaning the same Himalayan development that drives regional economic growth is also a documented contributor to the hazard it must then manage; landslide, cloudburst and GLOF risk in the Himalayas cannot be analysed independent of this development-risk tension. **Boundary:** WRONG: this topic does not cover the physical mechanics of slope failure or glacial-lake formation in geomorphological detail (Geography's domain); it covers risk governance built on that physical base.

## 2. Risk-system analysis



**Analytical claim:** FACT: VisionIAS explicitly names anthropogenic radiative-balance alteration - "mass tourism," "roads and hydropower projects" - as a GLOF accelerant in the Himalayan region "resulted in faster glacial melting and resultant GLOFs" (PDF p. 79). An advanced answer should treat this as direct evidence for the risk-creation-through-development thesis (topic 01's advanced file), applied specifically to the Himalayan cryosphere.

## 3. Legal/institutional depth

- FACT: **NDMA's explicit non-issuance of cloudburst-specific guidelines**

(PDF p. 33) is itself a governance gap worth analysing: cloudbursts are managed only by extension of flood and landslide guidelines, meaning no cloudburst-specific institutional accountability, monitoring metric or dedicated budget line exists in the source's own account - an advanced answer should name this absence explicitly rather than assume cloudburst management is institutionally equivalent to flood/landslide management.

- FACT: **State-level responsibility for area-specific measures:** landslide

mitigation explicitly calls for "area-specific measures" and land-use restriction "in the high vulnerability zones" (PDF pp. 32-33) - an advanced answer should note this again places the enforcement burden at the State/local level, structurally identical to the earthquake retrofitting-responsibility pattern (topic 05).

- ANALYSIS: **\*\*Avalanche warning sits with a defence-research institution, not**

NDMA/GSI\*\*: DGRE (Defence Geoinformatics Research Establishment, DRDO) issues India's avalanche warnings/forecasts for the Himalayan/Ladakh belt using snowpack, slope and weather data - an advanced answer should note this is institutionally distinct from GSI's landslide-forecasting mandate and ISRO's glacial-lake monitoring role (Section 5), meaning "Himalayan mass-movement hazard governance" is actually split across at least three separate institutions by hazard type, not a single unified authority.

## 4. Evidence and Indian applications

- FACT: **Mining-linked landslides as a distinct sub-category:** VisionIAS's

own zonation explicitly separates general slope-instability landslides from "landslides due to mining and subsidence," naming Jharkhand, Odisha, Chhattisgarh, MP, Maharashtra, AP, Karnataka, TN, Goa and Kerala (PDF p. 32) - an advanced answer should use this precise list as the evidentiary basis for the 2025 adjacent-only mining-environmental-hazard cross-link, rather than a generic "mining causes landslides" statement.

- FACT: **\*\*The Uttarakhand 2013 disaster as a documented multi-cloudburst,**

multi-hazard event\*\*: VisionIAS notes it "had several cloudburst events, two of them big ones" (PDF p. 33) - a concrete example an advanced answer can use to demonstrate that a single named disaster event frequently involves multiple simultaneous/cascading hazard mechanisms (cloudburst, flash flood, landslide), not one isolated hazard.

## 5. Technology/data and warning limits

- ANALYSIS: **Cloudburst's near-irreducible prediction gap:** VisionIAS states

prediction is "very difficult... due to its very small scale in space and time" (PDF p. 35), despite IMD's real-time rainfall monitoring - an advanced answer should distinguish this from earthquake's *complete* unpredictability (topic 05): cloudburst is difficult but not categorically impossible to forecast at short lead times, whereas earthquake prediction is denied outright by the same source; treating the two as equally unpredictable is imprecise.

- **FACT: ISRO's Himalayan glacial-lake monitoring role** (PDF p. 79) is the

correct institutional anchor for any claim about GLOF-risk remote- sensing capability; **CURRENT:** current monitoring-coverage claims should be verified against ISRO's and GSI's current published capability rather than assumed comprehensive.

- **ANALYSIS:** **\*\*Avalanche prediction shares cloudburst's short-lead-time**

character, but with a different physical basis\*\*: DGRE's warnings are built on snowpack-stability, slope and weather-forecast data rather than rainfall intensity - an advanced answer should not conflate avalanche forecasting with cloudburst prediction (Section 5 above) or earthquake's *complete* unpredictability (topic 05); each hazard's prediction/warning capability rests on a distinct data foundation and should be assessed separately.

## 6. Vulnerability, equity and community lens

- **ANALYSIS:** Himalayan/hill-state populations dependent on tourism and

hydropower-linked employment face a structural tension: the same economic activities sustaining their livelihoods are documented accelerants of the hazard threatening them (Section 2) - an advanced answer should name this as a genuine equity/livelihood dilemma, not a simple "stop development" recommendation, since alternative livelihood options are not detailed in the source material.

## 7. Prevention/mitigation/response/recovery trade-offs

- **ANALYSIS:** **\*\*Hydropower/infrastructure development versus glacial-risk**

containment is the central Himalayan trade-off.\*\* VisionIAS names hydropower projects among GLOF accelerants (Section 2) while elsewhere in the broader source material (topic 16, 18) infrastructure investment is treated as a development priority - an advanced answer should name this direct tension explicitly, recommending risk-informed siting/design (e.g. avoiding glacial-lake-proximate hydropower siting) rather than either blanket development or blanket restriction.

- **ANALYSIS:** **\*\*Terrace farming versus Jhumming (shifting cultivation) in the**

North-East\*\*: VisionIAS recommends terrace farming replace Jhumming to reduce landslide risk (PDF p. 33) - an advanced answer should note this recommendation has social and agronomic implications (shifting cultivation is often linked to specific community land-use practices) not detailed in the source, meaning implementation requires community consultation, not top-down imposition.

## 8. Governance and implementation gaps

- **WRONG:** **\*\*The "landslide vulnerability zones are mapped, therefore risk is**

managed" fallacy.\*\* VisionIAS's own existing-challenges list - integrating landslide concerns into DM plans across all levels, moving from "piecemeal remediation" to "simultaneous and holistic implementation," establishing a techno-legal regime, enforcing "zero tolerance against deliberate environmental violence," and building a national/international disaster-knowledge network (PDF p. 33) - shows mapping alone does not close the enforcement or coordination gap.

- **ANALYSIS:** **Cascading-hazard planning gap:** because earthquake, landslide, cloudburst and GLOF risks are causally linked in the Himalayan context (Section 2), single-hazard planning (e.g. an earthquake plan that ignores landslide-triggering) under-addresses actual risk - an advanced answer should recommend integrated, cascade-aware Himalayan risk planning explicitly.

## 9. Financing/monitoring/accountability

- **ANALYSIS:** VisionIAS does not detail landslide/GLOF-specific financing

instruments beyond the general call for "landslide insurance and compensation for losses" (PDF p. 33); any current claim about a specific insurance/compensation scheme requires a dated NDMA/State source.

- **CURRENT:** GLOF-risk project screening (e.g. environmental clearance conditions for Himalayan hydropower) should be verified against current MoEFCC/State-government project-appraisal practice, not assumed from this source alone.

## 10. CURRENT: Current official anchor and freshness protocol

- **CURRENT:** **\*\*GSI's National Landslide Forecasting Centre and the Bhusanket/**

Bhooskhalan platforms **are the correct current anchor for landslide-monitoring and forecasting-service status: NLFC inaugurated 19 July 2024; Regional Landslide Forecasting System covering 21 districts - 4 operational, 17 experimental - as of December 2025; National Landslide Susceptibility Mapping completed over ~4.3 lakh sq km in 19 States/UTs\*\*.**

- ANALYSIS: \*\*The operational/experimental split is the analytically important

fact, not the district count.\*\* Only 4 of 21 districts had operational forecasts; the remaining 17 were experimental. An advanced answer should therefore avoid the "India now has landslide early warning" formulation and say instead that India has moved from *susceptibility mapping* (a static, where-question answered over 4.3 lakh sq km) to *forecasting* (a dynamic, when-question answered operationally in four districts) - the gap between the two being the real measure of capability.

- CURRENT: For **GLOF**, the current anchors are NDMA's \*\*GLOF Guidelines

(October 2020) **and the National GLOF Risk Mitigation Programme (Rs 150 crore: Rs 135 crore NDMF + Rs 15 crore State; Arunachal Pradesh, Himachal Pradesh, Sikkim, Uttarakhand), against 195 high-risk lakes officially listed as of 4 December 2024. ANALYSIS: Compare the programme outlay with the number of high-risk lakes before describing coverage as adequate; and note the list is described as dynamic\*\***, so any lake count must carry its date.

- CURRENT: **DGRE's (DRDO) current avalanche-warning bulletins/services** are

the correct current anchor for any avalanche-warning-coverage or forecast-accuracy claim, distinct from GSI's landslide anchor above.

- WRONG: Do not present VisionIAS's "15% of land area" landslide estimate or its vulnerability-zone list as current without this verification.

## 12. Mains-ready framework

**Central thesis:** Himalayan landslide, cloudburst and GLOF risks are best analysed as a single, development-accelerated, cascading-hazard system - glacial retreat, slope destabilisation and extreme-rainfall events are each intensified by the same tourism/infrastructure development sustaining regional livelihoods - requiring risk-informed siting and integrated, cascade-aware planning rather than single-hazard management.

1. **Name the specific hazard and its precise mechanism** (Section 2 of the basic file), avoiding treating landslide/cloudburst/GLOF as interchangeable.
2. **Show the development-acceleration link** (Section 2 above) with the specific GLOF-accelérant evidence (mass tourism, roads, hydropower).
3. **Name the cascading-hazard sequence** relevant to the specific question (earthquake-landslide-outburst; cloudburst-flash-flood- landslide).
4. **Cite the mining-linked-landslide State list** (Section 4) precisely when cross-linking to the 2025 adjacent mining question.
5. \*\*Flag the hydropower-versus-glacial-risk and Jhumming-versus-terrace-farming trade-offs\*\* (Section 7) as genuine, community- sensitive policy choices.
6. **Close by relating Himalayan risk to risk-informed development** (topic 01) and climate-linked glacial retreat (topic 15).

## 13. Study links

- FACT: Foundation companion:  
basic/10\_Landslides-Avalanches-and-GLOF-Risk.md.
- FACT: 00\_Master-Framework.md Sections 5, 7 and 8 - the geological-hazard grouping, PYQ application box and recurring analytical gaps (single-hazard-silo gap) reused across this folder.
- ANALYSIS: **Downstream topics in this folder:** Topic 05 develops seismic-

landslide linkage; topic 04 develops multi-hazard early warning generally; topic 15 develops climate-linked glacial retreat and adaptation in full.

## CONSOLIDATED REGISTER NOTES

### Landslides, Avalanches and GLOF Risk: LANDSLIDE ROCKFALL DEBRIS-FLOW AVALANCHE GLOF AND RISK MAP

1. **Landslide:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.
2. **Rockfall:** Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow.
3. **Debris flow:** A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.
4. **Avalanche:** An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.
5. **GLOF:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.
6. **Susceptibility versus hazard:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.
7. **Risk:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.
8. **Conditioning factors:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.
9. **Triggers:** Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.
10. **Cascading risk:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.
11. **GSI role:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.
12. **DGRE role:** DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.
13. **NRSC-ISRO role:** NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.
14. **Glacial-lake monitoring:** Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.
15. **Zoning and land use:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.
16. **Slope and drainage measures:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope.
17. **Route and settlement planning:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

18. **Warning and evacuation:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.
19. **Institutional coordination:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.
20. **Technology-outcome firewall:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

## **Landslides, Avalanches and GLOF Risk: SUSCEPTIBILITY FORECAST INSTITUTION TECHNOLOGY AND OUTCOME FIREWALLS**

- Do not treat landslide, rockfall, debris flow, avalanche and GLOF as synonyms.
- Do not confuse intense rainfall or a cloudburst with the mass movement it may trigger.
- Do not equate susceptibility, hazard and risk maps.
- Do not assign GSI's landslide role to DGRE or DGRE's avalanche role to ISRO.
- Do not convert experimental forecasting into nationwide operational warning.
- Do not describe remote sensing as direct prediction of every failure.
- Do not assume retaining walls or drainage make an unsuitable site safe.
- Do not recommend road closure or evacuation without alternative access and safe destinations.
- Do not use a dynamic glacial-lake list or programme figure without its date and owner.
- Do not infer safe outcomes from a map, bulletin, sensor, scheme or structure.

## **Landslides, Avalanches and GLOF Risk: ZONING DRAINAGE SLOPE ROUTE SETTLEMENT WARNING EVACUATION SPINE**

IDENTIFY THE MOVING MATERIAL AND HAZARD PROCESS

- > SEPARATE CONDITIONING FACTORS FROM EVENT TRIGGERS
- > DISTINGUISH SUSCEPTIBILITY HAZARD EXPOSURE AND RISK
- > TRACE LANDSLIDE BLOCKAGE FLOOD AND GLOF CASCADES
- > ASSIGN GSI DGRE NRSC ISRO CWC STATE DISTRICT AND OPERATOR ROLES
- > COMBINE ZONING DRAINAGE SITE WORKS ROUTE AND SETTLEMENT PLANNING
- > CONNECT WARNING TO CLOSURE EVACUATION SAFE DESTINATION AND VERIFIED OUTCOME

## **Landslides, Avalanches and GLOF Risk: CURRENT GSI DGRE NRSC ISRO NDMA AND UNCERTAINTY BOUNDARY**

Official attempts covered GSI Bhusanket roles, a dated DGRE avalanche bulletin, NRSC's Landslide Atlas, ISRO's atlas route and NDMA pages. Thin, raw-PDF and transport-failed pages are logged; no current hazard level, forecast accuracy, lake count, route status, casualty, event attribution or operational-readiness claim was invented.

## **COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM**

**High-yield use:** Follow the panels as one topic-specific conceptual atlas: root question, classifications, processes or arguments, comparisons, objections and a final answer-writing spine. This text-native master is separate from both graphical flowchart artifacts.

### **ASCII MASTER FLOW - PANEL 1/12: Mountain-hazard taxonomy**

LANDSLIDE -> rock debris or earth movement  
ROCKFALL -> detached rock falls / bounces / rolls  
DEBRIS FLOW -> water-rich channelised mass  
AVALANCHE / GLOF -> snowpack release / lake outburst flood

### **ASCII MASTER FLOW - PANEL 2/12: Susceptibility-hazard-risk ladder**

SUSCEPTIBILITY -> WHERE terrain may fail  
HAZARD -> probability timing intensity  
EXPOSURE + VULNERABILITY + CAPACITY  
RISK -> expected harm and disruption

### **ASCII MASTER FLOW - PANEL 3/12: Slope-failure mechanism**

CONDITIONING -> geology slope weathering drainage land use  
TRIGGER -> rain quake erosion cutting loading  
FAILURE -> slide fall or flow  
EXPOSURE -> route settlement service consequence

### **ASCII MASTER FLOW - PANEL 4/12: Cascade map**

EARTHQUAKE / RAIN -> LANDSLIDE  
LANDSLIDE -> RIVER BLOCKAGE  
BLOCKAGE OR LAKE-DAM FAILURE -> OUTBURST FLOOD  
CLOUD BURST -> FLASH FLOW + DEBRIS + SLOPE FAILURE

### **ASCII MASTER FLOW - PANEL 5/12: Institutional owner matrix**

GSI -> landslide maps inventory research forecast development  
DGRE / DRDO -> snow and avalanche warnings  
NRSC / ISRO -> Earth observation and inventories  
CWC / STATE / DISTRICT -> downstream water and public action

### **ASCII MASTER FLOW - PANEL 6/12: Glacial-lake monitoring rail**

- 1 REPEAT SATELLITE SCREENING
- 2 LAKE DAM SLOPE ICE AND OUTFLOW CHARACTER
- 3 FIELD VALIDATION + DOWNSTREAM EXPOSURE
- 4 WARNING EVACUATION AND REVIEW

### **ASCII MASTER FLOW - PANEL 7/12: Zonation-to-decision bridge**

HIGHER SUSCEPTIBILITY / HAZARD?  
AVOID OR REQUIRE SITE INVESTIGATION  
CONTROL LAND USE DRAINAGE CUTTING LOADING  
MAP ROUTE CLOSURE EVACUATION AND RESIDUAL RISK

### **ASCII MASTER FLOW - PANEL 8/12: Slope-measure portfolio**

WATER -> surface / subsurface drainage  
GEOMETRY -> controlled excavation and toe protection  
MATERIAL -> retaining protection bioengineering  
LIMIT -> site-specific design maintenance residual risk

### **ASCII MASTER FLOW - PANEL 9/12: Route and settlement resilience**

ALIGNMENT / SITING + SPOIL / DRAINAGE CONTROL  
ALTERNATIVE ACCESS + SAFE STOPPAGE / CLOSURE  
ASSEMBLY AREA + EVACUATION ROUTE  
HOSPITAL POWER TELECOM WATER CONTINUITY

### **ASCII MASTER FLOW - PANEL 10/12: Warning-to-action chain**

MONITOR -> ASSESS -> FORECAST / BULLETIN  
AUTHORISE -> COMMUNICATE UNCERTAINTY  
CLOSE ROUTE / EVACUATE -> SAFE DESTINATION  
VERIFY RECEIPT ACTION AND FEEDBACK

### **ASCII MASTER FLOW - PANEL 11/12: Development-risk balance**

CONNECTIVITY TOURISM HYDROPOWER LIVELIHOODS  
VERSUS  
SLOPE DRAINAGE GLACIAL-LAKE AND DOWNSTREAM RISK  
VERDICT -> RISK-INFORMED SITING NOT BLANKET CLAIMS

### **ASCII MASTER FLOW - PANEL 12/12: Mountain-risk answer spine**

CLASSIFY PROCESS -> SEPARATE CONDITION FROM TRIGGER  
DISTINGUISH SUSCEPTIBILITY HAZARD AND RISK  
ASSIGN GSI DGRE NRSC CWC AND LOCAL ACTION  
COMBINE ZONING DRAINAGE ROUTES WARNING EVACUATION + LIMITS